Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10	(a)	(i)		temporary hard water (1) award (1) for any of following • same volume of soap solution needed (to produce a lather) after boiling as sample **C** / soft water • same volume of soap solution needed (to produce a lather) after boiling as after adding washing soda • it is soft after boiling / hardness is removed by boiling neutral answers - less soap needed after boiling hardness decreased after boiling			2	2		2
		(ii)		**D** (1) award (1) for either of following • less soap solution needed (to produce a lather) after boiling but still more than needed for sample **C** / soft water • boiling did not remove all the hardness / partly softened by boiling • less soap needed after boiling and less again after washing soda			2	2		2

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			award (1) for similarity in results e.g. • (both groups can draw the) same conclusions for samples **A**, **C** and **D** • volumes in the second experiment are approximately three quarters the values in the first experiment for samples **A**, **C** and **D** • results for samples **A**, **C** and **D** are reproducible award (1) for different conclusions drawn for sample **B** e.g. • first group found sample **B** to be permanently hard water but the second group found that is contains some temporary hardness and some permanent hardness			2	2		2
	(c)			calcium / magnesium **ions** <u>swap places with</u> / replaced by / exchanged for sodium **ions** (2) there must be one reference to <u>ions</u> if not complete answer award (1) for either of following calcium / magnesium **ions** removed / replaced sodium **ions** added if no mention of **ions** anywhere award (1) for calcium / magnesium swaps places with / replaced by / exchanged for sodium do not accept 'displace'	2			2		
	(d)	(i)		3.2×10^{-3} mol (2) award (1) for 0.0032		2		2	2	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		moles $MgCl_2 = \frac{0.384}{95} = 4.04 \times 10^{-3}$ mol more **Mg^{2+} ions** in second sample / magnesium chloride sample therefore it has more hardness (1) accept more moles of magnesium chloride than magnesium sulfate (in the same mass because M_r of magnesium chloride is smaller) ecf possible from part (d) (i)			2	2	1	
				Question 10 total	**2**	**2**	**8**	**12**	**3**	**6**